

The Utility Problem

The grid maximum capacity is very important to utilities because it provides information on whether an additional load can be connected at a given location. These values determine if connection requests will be accepted or rejected, thus these values might impact the overall satisfaction of the consumers in the distribution area.

In many utilities, the grid maximum capacity is traditionally assessed based on the sum of the contracted capacity. The contracted capacity might be defined in different ways for residential, commercial and industrial customers:

1. For residential (and small commercial) consumers it is usually defined by the value of the Master Circuit Breaker (MCB);
2. For large commercial and industrial consumers, it is defined in a service agreement as agreed & contracted capacity.

However, this value is often oversized, either historically, when the customer had no idea of his consumption, or because the customer did not compare this value (contracted capacity) with his real measured time-series data or for other reasons. These unnecessary high contracted capacity values impact the utility because the grid dimensioning and grid maximum capacity is based on these values. However, these values are almost never representative of the real world.

For these reasons it is important to analyze such occurrences to get better insights. If this type of analysis is conducted, the utility will know real situations in the areas of the grid in terms of contracted capacity vs. actual load. The utility will know where the spare capacity is higher or smaller. This could save large investment costs by reinforcing the grid only in places where it is strictly necessary.

The Data Solutions and Methods

To describe the methodology of this analysis in general terms, it is necessary to mention the required data sets which are used for this type of analysis. These datasets are:

1. Information about the master circuit breakers (contracted capacity) of all consumption points connected to the selected feeder/transformer.
2. Netload consumption time series data from all connected loads to the selected feeder/transformer.

After the collection of all required data and after the assurance that data are correct and synchronized, the process of getting the results of such analysis involves several steps. These steps are summing all load usage time series on the feeder/transformer and comparing this information with the sum of the master circuit breakers (contracted capacities) of all consumption points.

Actual vs. Contracted Grid Capacity Analysis

[UC17-01] - Awesense Energy Transition Platform Use Case



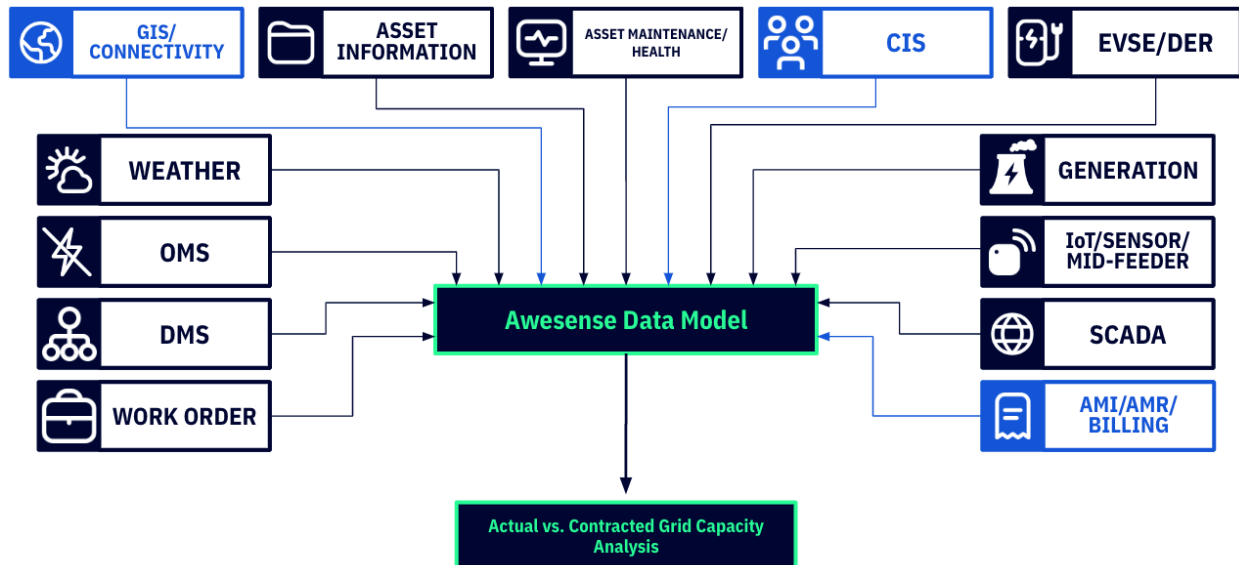
With Awesense Energy Transition Platform, conducting such analysis is significantly simplified. Awesense Energy Data Model's (EDM) data model and data access functions are the main aspects making such analysis easy to deliver.

1. The core dataset for the Awesense Energy Transition Platform, the GIS data, is modelled in a versatile data model which recognizes logical connectivity. In addition, the EDM's unique GIS querying functions allow analysts to simply obtain all connected assets to any point (all respecting connectivity and switchable element statuses). In this case, the function such as *grid_get_downstream()* and *ts_data_source_select()* would retrieve all connected meters on the feeder/transformer.
2. The consumption/generation or netload time series can be easily obtained as well due to EDM's built-in linkage of time series to the underlying GIS data. For analysts, a powerful function is available at their disposal. The result is a time series which can be turned to net load (if necessary) and then summed.

Completing this analysis is just a matter of setting up the algorithms for all consumption points on the feeder/transformer being analyzed, and then the kW sum data from the time series data and the capacity sum data are compared. This is calculated for all feeders/transformers and ranked according to the largest percentage difference. These results can be further used in Grid Planning (to find areas with large differences).

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. GitHub [link](#) to Jupyter notebook.